Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	5 (Term 1)	**Year:**	10–11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will distinguish modern, smart and composite materials through their behaviour and structure. They will develop their understanding through a teacher explanation, a worked example and an evidence-based classroom task. In the second lesson, students will select a material for a defined purpose using properties and constraints. Their recorded reasoning, practical evidence and final checks will show whether they can apply the ideas to a defined user, product or design problem.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher questioning, labelled diagrams, case comparisons and the completed independent tasks. Check accurate terminology, evidence linked to the user or brief and a reasoned explanation of choices. Review recorded tests or traces alongside conclusions, and use the final check in both lessons to identify the next skill needing practice.

LESSON SLOTS - TUESDAY AND WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	22 September 2026	By the end of this lesson, students will be able to distinguish modern, smart and composite materials through their behaviour and structure.	**H:** Show an image of a colour-changing strip and a layered sheet. Ask whether both are called smart for the same reason or whether they illustrate different material ideas. **ID:** Explain that smart materials respond to a stimulus such as temperature or light; modern materials are developed for improved properties; composites combine constituents to achieve useful properties. Discuss thermochromic pigment and a fibre-reinforced composite, noting that categories can overlap and each material has limitations. **WD:** Use the book examples to trace stimulus and response for a thermochromic indicator. Then label fibres and matrix in a composite diagram and explain their different roles. Model why	**Support:** Provide paired examples, a stimulus-response arrow diagram and a labelled composite outline. Rehearse the reason for one category together before students classify a new case. **Extension:** Students explain why a smart response may be useful but unsuitable for some conditions, and compare that limitation with a conventional alternative.

			naming a material without explaining its behaviour does not justify its selection. **YD:** Students sort illustrated material case cards, giving a reason for each classification. They annotate a stimulus-response example and a composite structure, then connect one property to a product use. All unfamiliar material behaviour is explored through images or teacher-prepared samples. **C:** Ask what makes the indicator smart and why a composite is more than a single homogeneous substance. Check classifications through explanations rather than accepting labels alone.	
2	23 Septemb er 2026	By the end of this lesson, students will be able to select a material for a defined purpose using properties and constraints.	**H:** Present a reusable protective case for a small electronic item. Ask which matters more: being very rigid, absorbing impact or allowing access, and why the answer depends on the brief. **ID:** Explain material selection through required properties, manufacturing method, availability, cost and environmental considerations. Distinguish strength, stiffness, toughness and hardness. Emphasise that a material family name does not establish the exact performance of every grade or product. **WD:** Compare supplied cards for card, a flexible polymer and a fibre-reinforced composite against a protective-case brief. Model excluding an option that fails a critical requirement, then comparing the remaining options and acknowledging cost or	**Support:** Offer a short glossary, pictures of the component and a partly completed matrix. Reduce the choices to two before asking students to justify the final selection independently. **Extension:** Students propose a combination of materials for different components and explain how joining or separation at end of life could complicate the design.

			repair limitations. **YD:** Students use a property matrix to choose a material for a defined product component. They cite supplied evidence for three requirements, explain one rejected option and suggest a simple test to check the most important property. Partners look for an unsupported claim or a conflict with the brief. **C:** Ask students to distinguish tough from hard using the proposed use. Check that the chosen material meets the essential requirements and that the suggested test measures the claimed property.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: 1.4 Modern and smart materials, composite materials and technical textiles
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared analysis worksheet, comparison grid or test-recording table
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?